**18** $(x - 8)(x + 4) = (x - a)^2 + b$ for all values of $x$.

Find the value of $a$ and the value of $b$.

$(x-8)(x+4)$

| × | $x$ | $4$ |
|---|---|---|
| $x$ | $x^2$ | $4x$ |
| $-8$ | $-8x$ | $-32$ |

$(x-2)^2$

| × | $x$ | $-2$ |
|---|---|---|
| $x$ | $x^2$ | $-2x$ |
| $-2$ | $-2x$ | $4$ |

**1 mark**
Expand original product

**1 mark**
Half−4 to get−2

$x^2 - 4x - 32 = (x - 2)^2 - 4 - 32.$

$= (x - 2)^2 - 36$, so $a=2$ and $b=-36$.

**1 mark**
a=2,b=−36

$a =$ .....

$b =$ .....

**(Total for Question 18 is 3 marks)**